SAL Engineering & Technical Institute

CE & CSE Department

Analysis and Design of Algorithms (BE04000241)

IMP Question Bank

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1. State the time complexity of following problems in descending order: Matrix multiplication, inserting element in binary search tree, inert node at beaning in linked list, linear search, selection sort, merge sort.
2. Summarize and explain the properties of algorithm
3. Apply bubble sort on the given data sequence: 77, 22, 55, 11, 33, 66, 44
4. Write the algorithm for binary search
5. Illustrate and solve recurrence for best and worst case of quick sort
6. Construct solution using Merge sort on the given data sequence: 88, 77, 22, 55, 11, 33, 66, 99, 44. Also state its recurrence and solve it
7. Define the characteristics of greedy algorithms
8. Explain activity selection problem.
9. Infer the Longest Common Subsequence for following strings: X = 10010100 Y = 10011
10. Explain the steps of greedy strategy for solving a problem
11. Explain Tower of Hanoi Problem, Derive its recursion equation and computer it's time complexity
12. Define the following terms: i) Function in mathematics ii) Linear inequalities
13. Write an algorithm for insertion sort. Calculate the best, average and worst case complexity of it.
14. The matrices A(5X10), B(10X15), C(15X20), and D(20X25) are given. Solve the matrix chain multiplication problem using dynamic programming.
15. For the given set of items and knapsack capacity = 5 kg, find the optimal solution for the 0/1 knapsack problem using dynamic programming.

    Item Weight Value I1 2 3 I2 3 4 I3 4 5 I4 5 6